

100 Days of Azure DevOps

Day 59 Handout — HPA scales pods; cluster autoscaler scales nodes; neither works without a metric

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

Architecture A — four levers that are not the same YAML object

Lever	What it changes	Requires
Fixed replicas	Deployment.spec.replicas	Honesty that load is flat. Fine for a lemonade stand
HPA	Pod replica count from a metric (CPU 70%, 1–5)	resources.requests.cpu on the container. metrics-server in kube-system
Cluster autoscaler	Node count when pods are Pending for capacity	Autoscaler add-on + a node pool that can scale. Not a substitute for HPA
Ingress	HTTP entry (ingress controller + Ingress object)	A controller (nginx/AGIC/etc). Not a Service type folklore
NetworkPolicy	Who may speak to whom in the CNI	A plugin that enforces (Azure Network Policy / Calico). Default-allow if none

Architecture B — pick HPA vs more replicas vs more nodes

Symptom	Wrong lever	Right lever
App CPU high, nodes fine	Add nodes 'because AKS'	HPA (or fixed replicas) after requests exist
Pods Pending: Insufficient cpu	HPA maxReplicas++ on a packed node	Cluster autoscaler / larger node / fewer requests
Feels slow, no metrics	HPA at 70% with requests: 0	Set requests; look at Azure Monitor Container Insights or kubectl top
Need HTTP routing	LoadBalancer per Service as architecture	Ingress survey; skip a CNI PhD if time-boxed
Lemonade stand	Copy a 12-object 'prod network' tutorial	Fixed 1 replica. Write why HPA is jewelry today

Architecture C — pitfalls that encode superstition in YAML

Pitfall	What you see	Fix
HPA without requests	CPU utilization 0% or unknown; no scale	resources.requests.cpu on the Deployment
HPA vs CA confusion	Pending pods; HPA already at max	Name the bottleneck: pod vs node
Ingress without controller	Ingress object exists; Address empty	Install/identify a controller first — or skip and write it
NetworkPolicy on default-deny by paste	Broke kube-system DNS / lost the app	Do not apply a random deny-all to kube-system. Survey only if time-boxed
Forgotten cluster	Day 56 gym membership still billing	Re-check destroy date. Monitoring does not pay the VM

One-liner to remember

Autoscaling without metrics is superstition with YAML. If you cannot name the signal, do not autoscale. HPA is pods; cluster autoscaler is nodes.

Step-by-step lab (personal Azure / Azure DevOps only)

Survey day. Personal notes over a CNI lab. If aks-lab still exists, the weekend destroy date is part of today's work.

- 1 Read HorizontalPodAutoscaler v2 and cluster autoscaler docs enough to write: HPA changes replicas; CA changes nodes; both need a signal.
- 2 Open your Deployment: confirm resources.requests.cpu exists (Day 55). If not, add 50m. Write why HPA is undefined without it.
- 3 Write docs/scale-day59.md with a table: when fixed replicas vs HPA vs CA. Include one sentence on Ingress vs LoadBalancer and one on NetworkPolicy default-allow.
- 4 Optional: apply the HPA sketch to kind/AKS and kubectl get hpa. Do not load-test a paid cluster just to see a replica flicker.
- 5 Skip deep CNI / AGIC labs if the clock says so — write what you skipped. Do not paste a default-deny NetworkPolicy into kube-system.
- 6 If AKS from Day 56 exists, confirm the destroy date is still on the calendar (or delete now). Container Insights is literacy, not a requirement to leave the cluster up.

Starter HPA — CPU 70% between 1 and 5, requests required

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: myapp
  namespace: workloads
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: myapp
  minReplicas: 1
  maxReplicas: 5
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 70
# Cluster autoscaler is a node-pool / add-on setting, not this object.
```

Done checklist

- Can separate HPA (pods) from cluster autoscaler (nodes) without mixing YAML objects
- Wrote when fixed replicas beat HPA for this app
- Know HPA needs resource requests and a metrics signal
- AKS destroy date re-checked or skip path still honest

Tomorrow — Day 60

Phase 6 mini project — one image in ACR, one automated run path